

# RAKESH VAIDEESWARAN MAHESH

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## EXPERIENCE

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### Amazon - Artificial General Intelligence (AGI Foundations)

*Applied Scientist II*

Sunnyvale, CA

Dec 2023 - Present

- Building Amazon Nova multimodal foundation models across pretraining, post-training (SFT, DPO, GRPO), reinforcement learning, and model evaluation.
- Developing reinforcement learning (RL) environments and gyms to improve factuality and grounding in agentic AI systems, including task design, dataset creation, reward formulation, and scalable evaluation.
- Contributed to the research, development, and release of **Nova 1 Lite, Pro** (Dec 2024), **Nova Premier** (Apr 2025), **Nova 2 Lite, Pro** (Dec 2025) foundation models, with a focus on text and multimodal understanding capabilities.
- Designed and implemented scalable data pipelines for curating high-quality training and validation datasets for LLM/VLM pretraining and post-training, driving improvements across model evaluation metrics.
- Identified model failure modes through systematic evaluation and error analysis, and developed targeted data and post-training interventions to improve reasoning, factuality, grounding, and multimodal understanding.
- Defined data requirements and quality criteria for third-party dataset acquisition, evaluating coverage, quality, and suitability for foundation-model pretraining and post-training.
- Collaborated across research, engineering, data, and product teams to translate experimental model improvements into production releases across the Amazon Nova family.

### Discover Financial Services - Account Takeover (ATO) Fraud Strategy

*Senior Data Science Analyst*

Chicago, IL

Feb 2023 - Dec 2023

- Developed data-driven fraud detection and prevention strategies for account takeover (ATO) and Zelle fraud, using Python, SQL, and Snowflake to identify emerging attack patterns.
- Led case reviews and deployed strategy updates that prevented hundreds of thousands of dollars in potential fraud losses.

### Amazon - Alexa AI

*Applied Scientist Intern*

Sunnyvale, CA

May 2022 - Aug 2022

- Developed a coarse-to-fine reasoning framework for Visual Question Answering (VQA), combining question representations with Faster R-CNN image features for multimodal reasoning.
- Built an explainable VQA system with Transformer/LSTM-based explanation generation, producing natural-language rationales on VQA-E and GQA-REX without degrading VQA accuracy; resulting work was published at the **NeurIPS 2022 TSRML Workshop**.

### Indian Institute of Science [SPIRE Lab]

*Junior Research Fellow (Guide: [Dr. Prasanta Kumar Ghosh](#))*

Bengaluru, India

Jul 2020 - Jul 2021

- Curated and validated multilingual speech datasets for the [MUCS 2021](#) Multilingual and Code-Switching ASR Challenge using Automatic Speech Recognition (ASR) metrics including Word Error Rate (WER) and acoustic likelihood.
- Trained a multilingual ASR baseline in Kaldi across six Indian languages, achieving ~30% Word Error Rate.
- Collaborated with Microsoft Research India, IBM, and Navana Tech on the MUCS 2021 challenge; resulting work was published at **Interspeech 2021**.

### University of New Brunswick

*MITACS Globalink Research Scholar (Guide: [Dr. Erik Scheme](#))*

Fredericton, Canada

May 2019 - Aug 2019

- Developed a deep learning system for air-writing sentence recognition using electromyography (EMG) and inertial measurement unit (IMU) sensor data from a wrist-worn device, using LSTM-based sequence modeling with CTC decoding.

### Indian Institute of Science

*Research Intern (Guide: [Dr. Neelesh Mehta](#))*

Bengaluru, India

May 2018 - Jul 2018

- Developed machine learning models to detect inlet air-nozzle blockages from simulated aircraft sensor data, with Support Vector Machines (SVMs) achieving the best performance at 86% accuracy.

## PUBLICATIONS & TECHNICAL REPORTS

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(\* denotes equal contribution)

### Peer-Reviewed Publications

1. **Stress tests REVEAL fragile temporal and visual grounding in video-language models** [[Link](#)]  
TV Sethuraman, Savya Khosla, Aditi Tiwari, Vidya Ganesh, Rakshana Jayaprakash, Aditya Jain, Vignesh Srinivasakumar, Onkar Kishor Susladkar, Srinidhi Sunkara, Aditya Shanmugham, Abbaas Alif Mohamed Nishar, [Rakesh Vaideeswaran Mahesh](#), Simon Jenni, Derek Hoiem  
*ICML 2026 Workshop on Mechanistic Interpretability | ICML 2026 Workshop on Combining Theory and Benchmarks*
2. **Can the decoded text from automatic speech recognition effectively detect spoken grammar errors?** [[Link](#)]  
Chowdam Venkata Thirumala Kumar, Meenakshi Sirigiraju, [Rakesh Vaideeswaran](#), Prasanta Kumar Ghosh, Chiranjeevi Yarra  
*SLaTE Workshop, Interspeech 2023*
3. **Towards Reasoning-Aware Explainable VQA** [[Link](#)]  
[Rakesh Vaideeswaran](#), Feng Gao, Abhinav Mathur, Govind Thattai  
*TSRML Workshop, NeurIPS 2022*
4. **A study on native American English speech recognition by Indian listeners with varying word familiarity level** [[Link](#)]  
Abhayjeet Singh, Achuth Rao, [Rakesh Vaideeswaran](#), Chiranjeevi Yarra, Prasanta Kumar Ghosh  
*O-COCOSDA 2021*
5. **Multilingual and code-switching ASR challenges for low resource Indian languages** [[Link](#)]  
Anuj Diwan\*, [Rakesh Vaideeswaran](#)\*, Sanket Shah\*, Ankita Singh\*, *et al.*  
*Interspeech 2021*
6. **Analysis of machine learning algorithms for wi-fi-based indoor positioning system** [[Link](#)]  
R Abishek\*, KR Abishek\*, N Hariharan\*, [Rakesh Vaideeswaran](#)\*, C Sundara Paripooranan\*  
*IMICPW 2019*

### Technical Reports & Model Cards

1. **Amazon Nova 2: Multimodal reasoning and generation models** [[Link](#)]  
Amazon Artificial General Intelligence (core contributor)  
*Amazon Science, 2025*
2. **Amazon Nova Premier: Technical report and model card** [[Link](#)]  
Amazon Artificial General Intelligence (core contributor)  
*Amazon Science, 2025*
3. **The Amazon Nova Family of Models: Technical Report and Model Card** [[Link](#)]  
Amazon Artificial General Intelligence (core contributor)  
*Amazon Science, 2024*

## PEER REVIEW & ACADEMIC SERVICE

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- Reviewer, **AAAI Conference on Artificial Intelligence** - 2026 (5 papers), 2027 (4 papers)
- Reviewer, **ACM Conference on Recommender Systems (RecSys)** - 2026 (2 papers)
- Reviewer, **ICML Workshops** - 2026 (9 papers)
- Reviewer, **Amazon ML Conference (AMLC)** - 2025 (2 papers)
- Lead Organizer, **Multilingual and Code-Switching ASR Challenge (MUCS workshop)** - 2021 (20+ submissions)

## TECHNICAL SKILLS

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- **Programming:** Python, C++, C, MATLAB, Bash, SQL
- **ML / LLM Frameworks:** PyTorch, Hugging Face Transformers, TensorFlow, scikit-learn, OpenCV, Kaldi, vLLM
- **Infrastructure:** Git, Snowflake, Kubernetes, AWS (S3, EKS, EMR)
- **Research Areas:** Large Language Models (LLMs), Vision-Language Models (VLMs), Natural Language Processing (NLP), Computer Vision, pretraining and post-training, Reinforcement Learning (RL), Supervised Fine-Tuning (SFT), multimodal learning, agentic AI, model evaluation, factuality and grounding.

## EDUCATION

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### University of Illinois, Urbana-Champaign

*Master of Engineering, Electrical and Computer Engineering*  
Focus Area: *Machine Learning and Signal Processing*

*Dec 2022*

**GPA:** 3.95/4

### National Institute of Technology, Tiruchirappalli (NIT Trichy)

*Bachelor of Technology, Electronics and Communication Engineering*  
Minor - *Computer Science and Engineering*

*Jul 2020*

**GPA:** 9.51/10

**Relevant Coursework:** Automatic Speech Recognition, Pattern Recognition, Random Processes, Computer Vision, Artificial Intelligence, Data Structures & Algorithms, Multimedia Signal Processing, Natural Language Processing

## PROJECTS

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### Dimension Estimation from Depth-Map of monocular Images [[Code](#), [Article](#)]

*Dec 2019 - Mar 2020*

- Trained a deep learning model based on Convolutional Autoencoder-Artificial Neural Network architecture to estimate the distance (in 3D space) between two arbitrary points on a 2D image, and achieved a Mean Deviation Error of 0.059 meters.

### Indoor Location Utility Service (ILocUS) [[Code](#), [Working Video](#)]

*Dec 2017 - Mar 2018*

- Implemented an indoor positioning system for indoor navigation using Wi-Fi, with signal values from 3 NodeMCU modules using k-Nearest Neighbors Algorithm (75% accuracy).
- Secured 1st place (among 16 teams) in a Product Ideation event organized by Qualcomm.

## EXTRACURRICULARS & LEADERSHIP

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- **Rubik's Puzzle Coach at XOOG (now [Famyo](#))** - One of the Top 4 coaches in May 2021 (among over 40 coaches); Mentored and coached over 20 kids aged 6-15 to solve the Rubik's Cube and motivated them to pursue advanced levels.
- Planned and Coordinated [Pragyan Cube Open 2019](#) (over 50 Participants) as the head of the organizing team, demonstrating leadership and teamwork; Organized [MUCS 2021 Workshop](#) (over 150 participants).